Chanho Lee

AI Scientist at LG CNS Future Robotics Lab

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Summary

AI Scientist at LG CNS Future Robotics Lab, specializing in VLA/VLM models for humanoid robot control. Full-cycle experience from simulation environment construction to real-robot deployment and manufacturing PoC. Expert in reinforcement learning, Sim-to-Real transfer, and efficient data pipelines.

Education

M.S. **Korea University**, Industrial and Management Engineering — Seoul, Republic of Korea, Mar 2021 – Aug 2023
- **GPA:** 4.19/4.5
- **Thesis:** Dynamic-Persistent CSMA: A Reinforcement Learning Approach for Multi-User Channel Access
- **Lab:** Supply Chain and Value Network Analysis Lab (Advisor: Prof. Taesu Cheong)
- **Visiting Scholar:** Drexel University, USA (Dec 2022 – Aug 2023, KIAT-funded)

B.S. **Korea University**, Mechanical Engineering (double major: Industrial & Management Engineering) — Seoul, Republic of Korea, Mar 2015 – Feb 2021
- **GPA:** 3.92/4.5

Experience

LG CNS, AI Scientist — Seoul, Republic of Korea, Dec 2024 – present, 1 year 6 months

Future Robotics Lab — Research on VLA-based humanoid robot solutions for manufacturing.
- IK+IL learning framework (independently developed): **95% pick-and-place success** on Unitree G1, reduced teleoperation dependency
- SAM2-based auto-labeling system: text input → automatic YOLO training data generation, eliminating manual labeling
- Real-robot deployment: Unitree G1 and H1-2 humanoid locomotion and manipulation policies
- VLA fine-tuning: GR00T, SmolVLA, ACT models adapted for manufacturing domain
- PoC clients: LG Chem, LG Energy Solution, Market Kurly, Daehan Steel
- Global collaboration: technical discussions with Skild, ARI Robotics (in English)

LG CNS, Vision AI Engineer — Seoul, Republic of Korea, Nov 2023 – Dec 2024, 1 year 2 months

Vision AI Business Team — Steel scrap grading AI for Daehan Steel.
- Accuracy **70% → 93%** (+23%p) by applying pixel-aligned structure from 3D reconstruction domain
- Optimized to 30-second processing per truck; fully deployed to production
- Joint venture established with client; system commercialized
- CLIP-based VLM experiments: scrap characteristic–image matching

Publications

Dynamic-Persistent CSMA: A Reinforcement Learning Approach for Multi-User Channel Access — Nov 2024

Chanho Lee, Seungkeun Park, Taesu Cheong

10.1109/ACCESS.2024.3506972 (IEEE Access)

Is View Planning Always Effective for 3D Reconstruction? Nov 2025
Chanho Lee, David Han
(Journal of Artificial Intelligence Research and Applications (JAIRA))

Awards

ETRI Spectrum Challenge Excellence Award (2nd consecutive win) Jan 2023
RL-based wireless channel access control competition, Electronics and Telecommunications Research Institute (ETRI)

ETRI Spectrum Challenge Excellence Award Jan 2022
RL-based wireless channel access control competition, Electronics and Telecommunications Research Institute (ETRI)

Skills

Programming: Python (Expert), C++ (Intermediate)

Deep Learning: PyTorch (Expert), TensorFlow (Intermediate)

Robotics & Simulation: Isaac Lab, MuJoCo, ROS | Unitree G1/H1-2/A2, Dexmate Vega, AgileX Piper

AI/ML: PPO, SAC, DAgger, Hierarchical RL, Imitation Learning, VLA (GR00T, SmolVLA, ACT), CLIP, SAM2, YOLO, PIFu

Tools: Git, LeRobot, BlenderProc, Teleoperation pipelines

Languages

Korean: Native

English: Advanced — TOEFL iBT 104 | 8-month research residency at Drexel University, USA